

with in-system programming (ISP) and in-application programming (IAP) features. It has 64 kB of Flash, 1024 bytes of internal RAM, 32 input/output lines, three 16-bit timers/counters, eight interrupt sources with four priority levels, serial peripheral interface (SPI) and enhanced UART, programmable counter array with pulse-width modulation and capture/compare functions, programmable watchdog timer, on-chip oscillator and clock circuitry. In addition, it supports two software-selectable power-saving modes—idle mode and power-down mode.

The system clock plays a significant role in the microcontroller's operation. A 11.0592MHz quartz crystal is connected across pins 18 and 19 of the microcontroller to provide it basic clock frequency. Capacitors C3 and C4 connected to the crystal maintain the resonance. Power-on reset at pin

9 is provided by the combination of capacitor C5 and resistor R2. Switch S1 is used for manual reset.

Port pins P3.0 and P3.1 of the microcontroller control IR transmitting LEDs TX1 and TX2 through transistors T5 and T4, respectively. Port pins P3.2 and P3.3 receive the detected signal from the collector of phototransistors T7 and T6, respectively. Phototransistors T6 and T7 conduct on receiving IR signals from TX2 and TX1 and provide a low output at pins P3.3 and P3.2 of the microcontroller, respectively.

Whenever any vehicle comes in between the IR transmitters and receivers, T6 and T7 stop conducting to provide a high signal to port pins P3.3 and P3.2 of the microcontroller, respectively, indicating the presence of a vehicle between the IR modules. The microcontroller counts the number of vehicles between IR transmitting LEDs

and photo-transistors by detecting the number of high signals at its pins P3.2 and P3.3. Traffic profile is computed by the microcontroller in a predefined delay (controller interval) according to the vehicle count.

The microcontroller controls red, yellow and green lights (connected to its pins P3.6, P3.5 and P3.4, respectively) according to the traffic profile. The traffic lights are connected to the microcontroller's output pins through relays. Transistor T3 drives relay RL3 for the red light. Similarly, yellow and green lights are controlled

by pins P3.5 and P3.4 through relays RL2 and RL1 driven by transistors T2 and T1, respectively. D5, D6 and D7 act as free-wheeling diodes for protec-

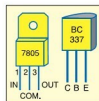


Fig. 3: Pin details of 7805 and BC337

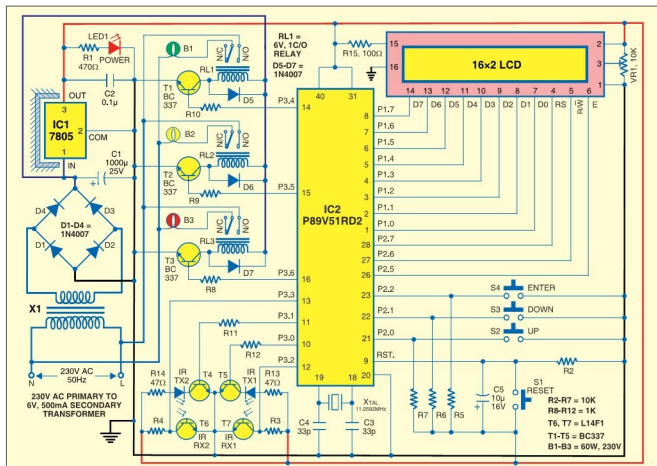


Fig. 2: Circuit diagram of traffic light controller